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Numerical study on weld pool behaviors and keyhole dynamics in magnetic-field-assisted laser-arc hybrid welding of aluminum alloy

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ABSTRACT

External magnetic field has been employed to improve appearance and suppress defects in the single laser or arc welding, and its physical mechanism in magnetic-field-assisted single laser or arc welding has been clearly clarified. However, few studies on the physical processes in magnetic-field-assisted laser-arc hybrid welding (MFA-LAHW) are conducted and its influencing mechanism is unclear. In this research, a comprehensive three-dimension numerical model is established to investigate the keyhole dynamics and weld pool behaviors in MFA-LAHW under the effect of different magnetic flux density. The results suggest that the interaction between a lower magnetic flux density (≤ 30 mT) and welding current is induced due to the existence of arc. Compared to the magnetic-field-assisted laser beam welding (MFA-LBW), this interaction induces a stronger Lorentz force up to the order of magnitude of 10^5 N/m³. The weld pool width is reduced while its length and height are increased because the Lorentz force facilitates the backward flow of thermal fluid, increasing the heat transfer from the keyhole and droplet to the rear weld pool edge. Moreover, the steady magnetic field improves the keyhole stability. The deform on keyhole rear wall brought by the droplet and vortex is suppressed, and keyhole size is widened under steady magnetic field, which reduced keyhole collapse. Furthermore, the comparative analysis with the case in MFA-LBW is carried out and the MFA-LAHW is verified to be more valuable for medium-thickness aluminum alloy. This work provides the underlying physics understanding of MFA-LAHW, which will broaden the application value of LAHW. © 2023 The Authors. Published by Elsevier B.V. This is an open access article under the CC

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1. Introduction

Laser-arc hybrid welding (LAHW) takes the advantages of the high welding efficiency and penetration depth in the laser

welding and the superior gap-filling ability in the arc welding, thereby compensates for the shortcomings of the two single welding methods [1]. Benefiting from interaction of laser and arc heat sources, LAHW has been developed to a promising technology to weld the medium-thickness plates [2]. However,

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the poor weld formation, inhomogeneous microstructure, and some porosity defects are usually found in the LAHW joint of aluminum alloys, deteriorating its estimate properties, which limits the industrial application of LAHW [3].

Magnetic-field-assisted welding technology has proven to be effective in eliminating defects and improving weld formation for various welding methods. For arc welding, the magnetic field was employed to obtain good weld appearance and improve joint quality by controlling motion of arc plasma [4]. Furthermore, the undercut and excessive sag defects were suppressed by producing electromagnetic force to change weld pool behaviors [5]. For the laser welding, the benefiting effects of external magnetic field have been verified by plenty of experimental works. Eliminating defects including hump defects [6], back sagging [7], weld porosity [8] with the magnetic-field-assisted technology in laser welded process were proposed in previous study. Additionally, the alternating magnetic field is also employed to homogenize elements [9,10], and refine microstructure [11,12] by stirring weld pool.

To figure out the effect of the external magnetic field on the laser welding process, some mathematical modelling works considering the magnetohydrodynamics (MHD) have been conducted to reveal the underlying physical phenomena involving keyhole dynamics and melts flows. Bachmann et al. [13–15] developed the numerical model with the keyhole morphology to study the weld pool flow behaviors in the MFA-LBW. Subsequently, more comprehensive multi-physics models using the volume of fluid method were established to deeply research the influence mechanism of external magnetic field on a series of beneficial effects of laser welding. Cao et al. [16] established a multi-physics model considering the keyhole dynamics to numerically explore the thermoelectric currents distribution in the

electromagnetic assisted laser welding. Similar models were considered to investigate the suppression mechanisms of root hump [17] and porosity defect [18] under steady magnetic field, and the element distribution homogenization mechanism under the alternating magnetic field [19,20]. To sum up, the fundamental physics of the MFA-LBW can be concluded as follows: A self-induced current is generated in the electrically conductive weld pool. Thereby, the Lorentz force is produced since the self-current is affected by the external magnetic field, which will change the weld pool flowing behaviors.

For LAHW, the welding current is a major component of the arc heat source. The Lorentz force is produced by the interaction between the welding current and the magnetic field, which differs from that in MFA-LBW. Thus, the physics mechanism of MFA-LAHW is more complicated. Relevant experimental studies have been conducted on MFA-LAHW, and its benefiting effects including grains refinement [21–23], and porosity suppression [24,25] have also been proposed, which improved our knowledge of MFA-LAHW technology. Nevertheless, the underlying physics understanding of weld pool behaviors in MFA-LAHW is still lacking.

Consequently, in present work, the effect mechanism of magnetic field on keyhole dynamics and weld pool behaviors in LAHW was revealed by setting up comprehensive three-dimensional multiphase transient model. The keyhole free surface was tracked through the volume of fluid method. The Lorentz forces and droplet under the magnetic field were analyzed. The profile evolution of the weld pool and keyhole stability under varying magnetic flux density were explained by analyzing thermal fluid flows. Moreover, the comparative analysis with the case in MFA-LBW was carried out to deepen the physics understanding of MFA-LAHW.

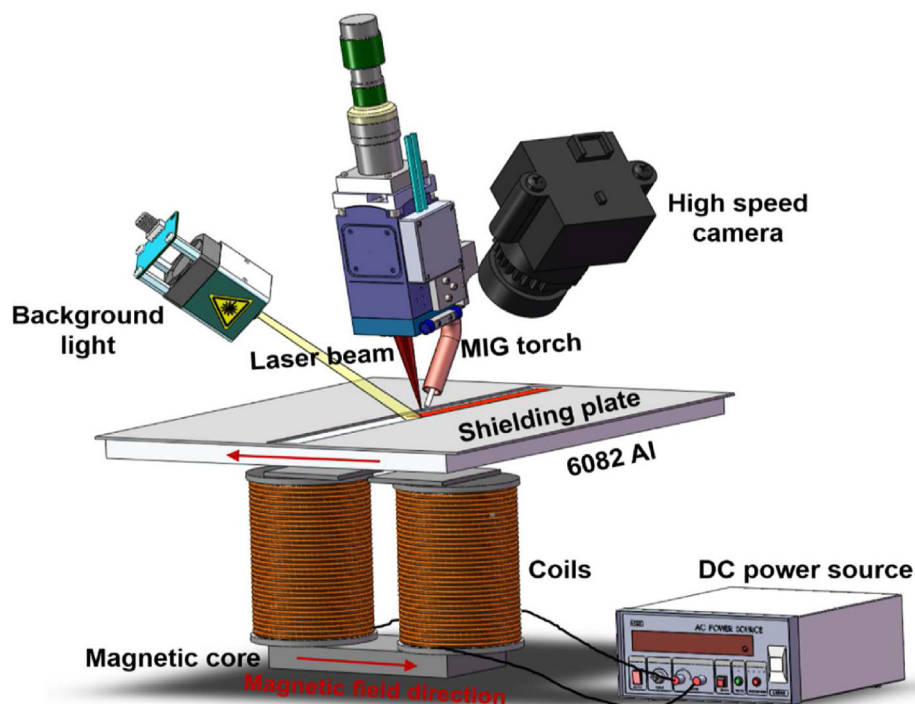


Fig. 1 – Schematic of the experimental platform.

2. Materials and experimental setup

The 6082 aluminum alloy samples with the relative magnetic permeability of one and high electrical conductivity is selected as base metal, which has a dimension of 100 mm × 50 mm × 8 mm. The oxide coating is removed through grinding and the surface is cleaned by acetone before welding.

The schematic of MFA-LAHW is shown in Fig. 1. The experiment is carried out with the assistance of a coupled system of arc welding system (Lincoln Power Wave R350) and the fiber laser system (IPG YLS-6000). Table 1 shows the relevant parameters in the experiment. The welding system is fixed and the metal moves with the help of a screw mechanism. The laser head is in lead and the arc torch is in trail. To reduce the negative influence of the magnetic field, a ferromagnetic steel shielding board is covered on the metal surface. The size of the steel is 2 mm thick and a hole with the width of 15 mm in the center area is set to the welding area.

The high-speed camera (IX I-SPEED 221) with an auxiliary light source is employed to collect the weld pool behavior. Two electromagnetic coils in series are employed to generate the magnetic field, and its parameters and arrangement have been reported in our previous paper [18]. The magnetic flux density in the welding area was set to 0, 15, and 30 mT as variable parameters. Moreover, the electromagnetic simulation method is adopted to research the distribution of magnetic flux density. A Gauss meter is utilized before welding to measure the magnetic flux as a validation procedure, which will be discussed in section 3.5.

3. Numerical models

This research focuses on the keyhole dynamics and weld pool behaviors in MFA-LAHW. The laser and arc heat source model are employed, and the driving forces applied on weld pool include self-electromagnetic force, recoil force, surface tension, buoyancy force, and arc pressure. Moreover, the numerical model introduces the Lorentz force in the weld pool. Table 2 summarizes all the nomenclatures. To simplify the model, some assumptions are given as follows.

Table 1 – Experimental parameters.

Process parameters	Value
Laser power (W)	4000
The laser beam parameter product (mm • rad)	8
Arc current (A)	210
Arc voltage (V)	24
Welding speed (m/min)	1.8
Wire feed speed (m/min)	10.6
Distance between wire tip and metal (mm)	2.5
Arc torch angle (°)	30
The angle between high-speed camera and metal surface (°)	75

- (1) The molten metal is an incompressible Newtonian fluid and the fluid flow is laminar.
- (2) The effects of laser-induced plasmas and shielding gas are not considered.
- (3) Ignoring the Joule heat and current generated by the magnetic field in the weld pool.
- (4) The formation and migration of laser-induced pores are not included in this model.

3.1. Governing equations

3.1.1. Mass equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = S_w \quad (1)$$

3.1.2. Momentum equation

$$\rho \left(\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v} \right) = -\nabla p + \mu \nabla^2 \vec{v} - \mu A \frac{(1 - f_L)^2}{f_L^3 + \delta} \vec{v} + \vec{S}_m \quad (2)$$

3.1.3. Energy equation

$$\rho \left[\frac{\partial h}{\partial t} + (\vec{v} \cdot \nabla) h \right] = \nabla \cdot (\omega \nabla T) + S_q \quad (3)$$

3.1.4. VOF equation

$$\frac{\partial F}{\partial t} + \nabla \cdot (\vec{v} F) = S_F \quad (4)$$

3.2. Heat source model

3.2.1. Arc heat source

Generally, an asymmetric double elliptical heat model is used to obtain the distribution of arc heat source in the single arc welding [26]. Given that the arc torch is inclined in LAHW, and the heat density of arc is affected by the distance between its position and the wire tip [27]. The heat flux decreases inversely correlated to the square of the distance between the position and the wire tip increases. Thus, the model of arc heat is optimized by a dimensionless factor (d), which is given as follows:

$$d = \frac{L_{pos}^2}{L_{arc}^2} \quad (5)$$

where, L_{pos} represents the distance between the wire tip and the weld pool surface, L_{arc} represents the length of arc. An improved heat model is proposed in Eq. (6).

$$q_a(x, y) = \begin{cases} \frac{6Q_a K_d}{\pi(a_f + a_r)bd} \exp\left(-\frac{3(x - d_{la})^2}{a_f^2} - \frac{3y^2}{b^2}\right) & x - d_{la} \geq 0 \\ \frac{6Q_a K_d}{\pi(a_f + a_r)bd} \exp\left(-\frac{3(x - d_{la})^2}{a_r^2} - \frac{3y^2}{b^2}\right) & x - d_{la} \leq 0 \end{cases} \quad (6)$$

where, a_f , a_r , and b are arc energy distribution coefficients. Q_a expressed as Eq. (7) is the arc heat input absorbed by the weld

pool, which equals to the total heat input of arc minus heat input melting wire filler. K_d is an adjustment parameter and it equals to the ratio between the theoretical and the actual value of the heat arc input exerted in the computational domain.

$$Q_a = \varphi UI - \rho h(\pi v_{\text{wire}} r_{\text{wire}}^2) \quad (7)$$

3.2.2. Laser heat source

The laser energy is represented by the Gauss-like distribution heat source model, and Eq. (8) shows its distribution equation. Moreover, the multiple reflection of laser beam is considered, which have been introduced in our previous study [18].

$$q_l(x, y) = \frac{P_l}{\pi R^2} \exp\left(-\frac{x^2 + y^2}{R^2}\right) \quad (8)$$

3.3. Driving force model

3.3.1. Arc pressure

The impingement of the weld pool by the plasma is crucial in the formation of the arc pressure. Eq. (9) expresses the distribution of arc pressure, which is determined by the welding current density [28].

$$P_a = \begin{cases} C_j \frac{3u_0 I^2}{\pi^2 (a_f + a_r) 2bd} \exp\left(-\frac{3(x - d_{la})^2}{a_f^2} - \frac{3y^2}{b^2}\right) & x - d_{la} \geq 0 \\ C_j \frac{3u_0 I^2}{\pi^2 (a_f + a_r) 2bd} \exp\left(-\frac{3(x - d_{la})^2}{a_r^2} - \frac{3y^2}{b^2}\right) & x - d_{la} \leq 0 \end{cases} \quad (9)$$

3.3.2. Arc shear stress

A double-ellipse arc shear stress distribution is employed in LAHW. The expressions of arc shear stress on the weld pool surface are expressed as follows [29]:

$$\tau_a(x, y, z) = \tau_{\max} g_a \left\{ 4.05 I^{-0.267} \left[\left(\frac{R}{d^2} - \frac{1}{d} \right) r + 1 \right] r \right\} \quad (10)$$

$$g_a(r) = \begin{cases} -0.363r^3 + 0.034r^2 + 1.322r - 0.0128 & (r \leq r_d) \\ 3.46 \exp\left(-\frac{r}{1.02}\right) + 0.0345 & (r > r_d) \end{cases} \quad (11)$$

where, $r = \sqrt{(x - v_w t - d_{la})^2 + y^2}$. The maximum arc shear stress ($\tau_{\max}$) is specified as 80 Pa based on the research of Campbell et al. [30].

The transient evolution of arc shear stress ($\vec{\tau}_a$) is calculated as Eq. (12).

$$\begin{cases} \vec{\tau}_a \cdot \vec{n} = 0 \\ |\vec{\tau}_a| = \tau_a(x, y, z) \\ \vec{\tau}_a = D \vec{\sigma} + E \vec{n} \end{cases} \quad (12)$$

where, D and E are both the real numbers.

3.3.3. Self-electromagnetic force

The self-electromagnetic force (F_e) that can drive the flow of the melt is generated by the current line of the weld pool, which is expressed as Eq. (13) [31]. where, σ_j is the distribution coefficient of electromagnetic force. It is obtained as follows: $\sigma_j = \sqrt{\frac{(a_f + a_r)b}{12}}$.

Table 2 – All the nomenclatures of this model.

Symbol	Nomenclature	Symbol	Nomenclature
$\vec{v}$	Velocity vector	m_w	Mass source from filler wire
P_0	Ambient pressure	K	Curvature of free surface
T_b	Boiling temperature	U	Voltage
δ	Small positive constant	β	Coefficient of thermal expansion
p	Pressure	P_l	Laser power
μ	Dynamic viscosity	ω	Thermal conductivity
ρ	Material density	$\vec{n}$	Normal vector along the surface
r_H	Effective radius of arc on baser metal	μ_0	permeability of vacuum
R	Universal gas constant	$\vec{g}$	Gravitational acceleration
F	Volume fraction	T_L	Liquid temperature
ϵ	Coefficient determined by the laser type and material properties	ΔH	Fusion latent heat of filler wire
L_v	Latent heat of evaporation	T_m	Melting temperature
d_{la}	Distance from wire tip to laser beam	T_0	Reference temperature
μ_m	Magnetic permeability	ρ_l	Liquid density
ω_c	Convective heat transfer coefficient	h	Enthalpy
v_{wire}	Wire feed speed	A_s	Temperature coefficient of surface tension
S_q	Energy source term	V	Free surface recession speed
δ_0	Surface tension coefficient	L	Thickness of base metal
C_j	Arc pressure adjustment factor	r_{wire}	Wire radius
ϵ	Thermal emissivity	I	Current
$\vec{S}_m$	Momentum source term	v_g	Vapor velocity
R	Laser beam radius at the focused plane	f_L	Liquid fraction
Re	Reynolds number	θ	Angle of arc torch and vertical line
σ	Stephan–Boltzmann constant	$\vec{B}_0$	External magnetic flux density
S_F	Volume source from the filler wire	T	Temperature

$$\begin{cases} F_{ex} = -\frac{u_m I^2}{4\pi^2 \sigma_j^2 r} \exp\left(-\frac{r^2}{2\sigma_j^2}\right) \left[1 - \exp\left(-\frac{r^2}{2\sigma_j^2}\right)\right] \left(1 - \frac{z}{L}\right)^2 \frac{(x - v_w t - d_{la})}{r} \\ F_{ey} = -\frac{u_m I^2}{4\pi^2 \sigma_j^2 r} \exp\left(-\frac{r^2}{2\sigma_j^2}\right) \left[1 - \exp\left(-\frac{r^2}{2\sigma_j^2}\right)\right] \left(1 - \frac{z}{L}\right)^2 \frac{y}{r} \\ F_{ez} = \frac{u_m I^2}{4\pi^2 \sigma_j^2 r} \left[1 - \exp\left(-\frac{r^2}{2\sigma_j^2}\right)\right]^2 \left(1 - \frac{z}{L}\right) \end{cases} \quad (13)$$

3.3.4. Surface tension

The continuous surface stress method is employed to deal with the surface tension (P_s) and its expression is given by Eq. (14) [32].

$$P_s = K[\delta_0 + A_\delta(T - T_m)] \quad (14)$$

3.3.5. Recoil force

The recoil force (P_r) is calculated according to the Clausius-Clapeyron equation, and it is described as Eq. (15) [33].

$$P_r = 0.54P_0 \exp\left(L_v \frac{T - T_b}{RTT_b}\right) \quad (15)$$

3.3.6. Buoyancy

The thermal buoyancy (F_β) is added into the momentum equation, which can be given as follows [34]:

$$F_\beta = -\beta\rho(T - T_m)\vec{g} \quad (16)$$

3.4. Droplet transfer

With the melting of wire filler, the droplet transfers from the wire tip into the weld pool with a spray transfer mode periodically in LAHW. In this experimental, the inclination angle between torch and vertical line is 30°, and droplet transfers into the weld pool along the torch axis. Thus, the droplet is treated as the liquid phase and it enters the calculating domain periodically from the velocity inlet boundary in this simulation. Furthermore, the initial temperature of droplet, velocity, transfer frequency, flowing time within a cycle, and the inlet radius are set in Table 3 based on the research of Gao et al. [27].

3.5. Considering effect of external magnetic field

3.5.1. Distribution of Lorentz force

A finite element model through the software *Ansys Electronics Desktop 2021* is established to depict the magnetic flux density of the weld area. The geometries and dimensions of magnetic coils and base metal are kept same as that of the actual components. Fig. 2 depicts the vector and contour of the magnetic

flux density distribution. The vector of magnetic flux lines along the vertical centerline of the base metal is parallel to the Y axis, as suggested in Fig. 2(a). In addition, the magnetic flux density value is reduced as the coordinate scale of the Z-axis increases. Along the y direction, it is also decreased symmetrically from the centerline of the weld pool. Thus, a binary polynomial was proposed to fit the magnetic flux density, which is expressed as Eq. (17).

$$B_y = B_{y0}(-194363.3x^3 + 1246.8x^2 - 4.317x + 1308266.7z^3 + 13,316.6z^2 + 81.9z + 0.377) \quad (17)$$

where, the effective ranges of y and z are -25–25 mm and 0–10 mm, respectively.

A Gaussian distribution is utilized to distribute the current density of the weld pool, and the current density distribution is expressed in Eq. (18), which considers the variation owing to the arc deflection in the presence of the magnetic field [31,35].

$$\begin{cases} J_{cx} = \frac{I}{2\pi L} \left[1 - \exp\left(-3 \frac{(x - x_d)^2 + y^2}{r_H^2}\right)\right] \frac{x - x_d}{(x - x_d)^2 + y^2} \\ J_{cz} = \frac{3I}{\pi r_H^2} \exp\left(-3 \frac{(x - x_d)^2 + y^2}{r_H^2}\right) \left(1 - \frac{z}{L}\right) \end{cases} \quad (18)$$

where, J_{cx} and J_{cz} represent the current density component along the x direction and the z direction. x_d is the distance between the coordinate origin and the position where the droplet transferred into the weld pool. The Lorentz force component along the x direction (F_{Lx}) is produced through the J_{cz} under the magnetic field B_y . Similarly, the Lorentz force component (F_{Lz}) along the z direction is produced through the interaction between the B_y and the J_{cx} . Eq. (19) expresses these interactions.

$$\begin{cases} F_{Lx} = B_y \times J_{cz} \\ F_{Lz} = B_y \times J_{cx} \end{cases} \quad (19)$$

3.5.2. Effect of magnetic field on droplet transfer

To treat the distribution of current density in the droplet transfer process, a liquid metal steam model is introduced

Table 3 – Droplet parameters defined by in this simulation.

Parameters	temperature	velocity	transfer frequency	flowing time within a cycle	inlet radius
Value	1800 K	1.65 m/s	100 Hz	1.12 ms	0.7 mm

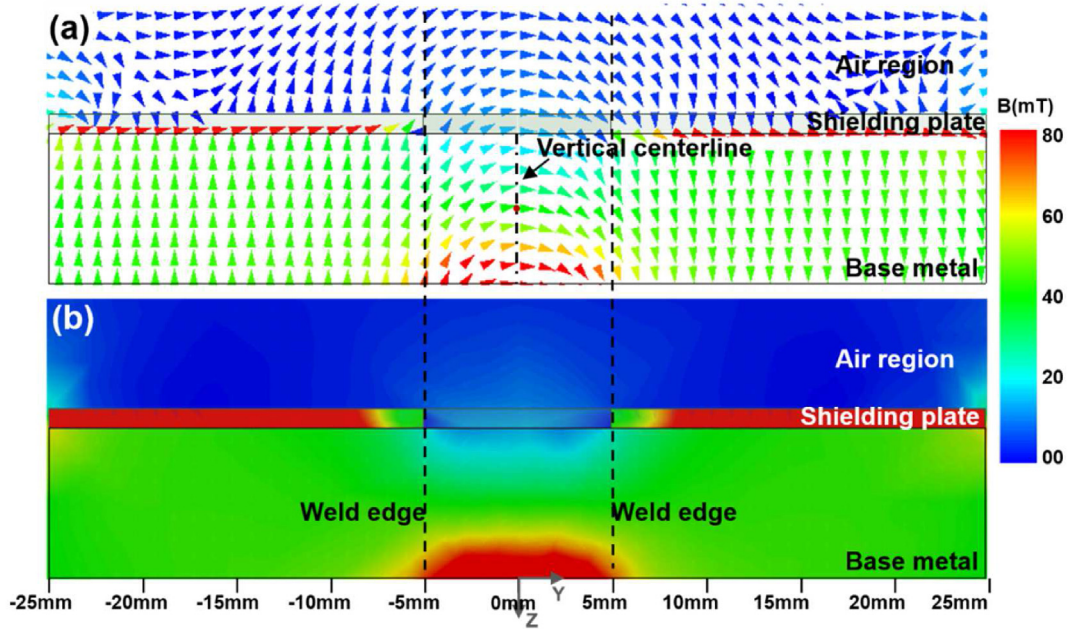


Fig. 2 – Vector and contour of magnetic flux density in Plane YZ: (a) vector; (b) contour.

based on the study of Li et al. [36]. When the liquid metal steam is produced at the wire tip, the welding current flows from the wire filler to the weld pool through the steam. Thus, the current density of the liquid metal steam can be given as Eq. (20) [37].

$$\begin{cases} J_{dx} = \left[\frac{I}{\pi r_{wire}^2} - \frac{(C_p \times \Delta T + \Delta H) \times \rho_w \times v_{wire}}{U} \right] \times \sin \theta \\ J_{dz} = \left[\frac{I}{\pi r_{wire}^2} - \frac{(C_p \times \Delta T + \Delta H) \times \rho_w \times v_{wire}}{U} \right] \times \cos \theta \end{cases} \quad (20)$$

where, J_{dx} and J_{dz} are the components of the current density along the x direction and z direction. Thus, the Lorentz force generated on the liquid metal steam is expressed as Eq. (21).

$$\begin{cases} F_{dx} = B_y \times J_{dz} \\ F_{dz} = B_y \times J_{dx} \end{cases} \quad (21)$$

3.6. Computational domains and boundary conditions

The size of the whole computational domain is $30 \times 18 \times 14 \text{ mm}^3$ and it is divided into the lower liquid domain and upper gas domain. Fig. 3 illustrates the boundary conditions and computational domains. A total of 945,000 uniform hexahedral grids with the size of $0.2 \times 0.2 \times 0.2 \text{ mm}^3$ are used. For the gas domain, the gas is set to argon with the pressure of 1 atm and the temperature of 300 K. To describe the addition of the droplets, the top surface is specified as the velocity input. The side surface is specified as pressure output. For the liquid domain, each of the surface is specified as wall.

The free surface of the gas and liquid phase were tracked through the VOF method to obtain the thermal boundary conditions, as given in Eq. (22):

$$k \frac{\partial T}{\partial n} + Q_a + Q_l = \omega(T - T_0) + \varepsilon \sigma (T^4 - T_0^4) + \rho_l V T_b \quad (22)$$

The velocity boundary conditions of free surface along the

normal direction and tangential direction are expressed as Eq. (23) and Eq. (24):

$$-p + 2\mu \frac{\partial v_n}{\partial n} = -P_a - P_d - P_r + \gamma_s K \quad (23)$$

$$\mu \frac{\partial v_n}{\partial n} = \frac{\partial \gamma_s}{\partial T} \frac{\partial T}{\partial s} \quad (24)$$

At the wall boundary, the velocities are set as zero and, Eq. (25) shows thermal boundary condition:

$$-k \frac{\partial T}{\partial n} = -\omega(T - T_0) - \varepsilon \sigma (T^4 - T_0^4) \quad (25)$$

3.7. Numerical implementations

In this research, the numerical model is established by the software ANAYS FLUENTS. The velocity and pressure are

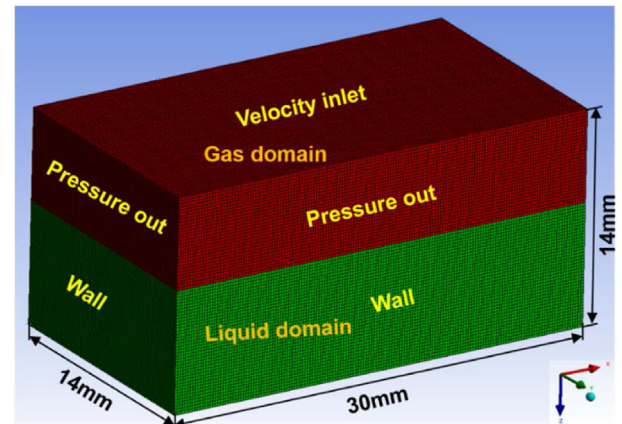


Fig. 3 – Boundary conditions and computational domains.

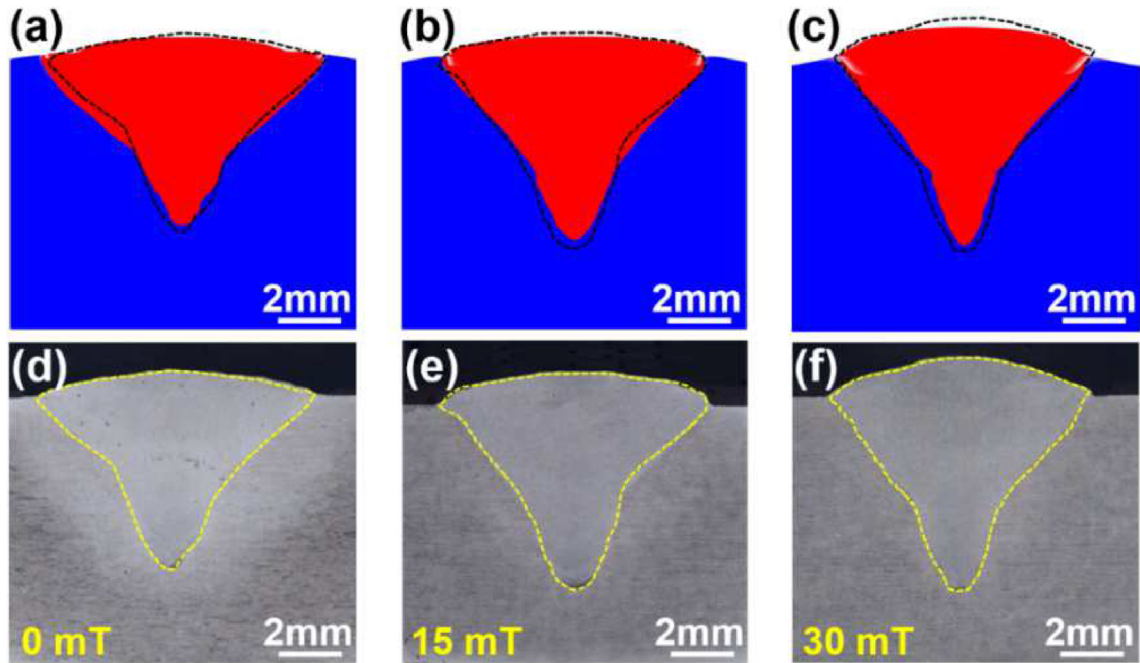


Fig. 4 – Comparison of cross-sectional morphologies: (a) (d) 0 mT; (b) (e) 15 mT; (c) (f) 30 mT.

coupled using the pressure-implicit with splitting of operator's algorithm and the governing equations are discretized using the second-order upwind method. The constant thermal-physical properties of the base metal can be found in our previous research [18]. The thermal conductivity, viscosity, and specific heat are displayed in Eqs. (26)–(28).

$$\mu = 10^{\left(\frac{576}{T} - 3.75\right)} \quad T \geq 930K \quad (26)$$

$$\omega = \begin{cases} -0.0317 \times T + 248.89 & T \leq 910K \\ -6.5 \times T + 6135 & 910K < T \leq 930K \\ 0.0339 \times T + 58.475 & T > 930K \end{cases} \quad (27)$$

$$C_p = \begin{cases} 0.2381 \times T + 833.33 & T \leq 910K \\ 7.35 \times T - 5638.5 & 910K < T \leq 930K \\ 1197 & T > 930K \end{cases} \quad (28)$$

4. Results and discussion

4.1. Verification of simulation models

The comparison of cross-sectional morphologies between experimental and numerical results under the variation of the magnetic flux density are depicted in Fig. 4. It reveals a good agreement between the predicted size and shape of cross-sectional morphologies with the experimental results. However, large deviations in the weld depth are observed, which

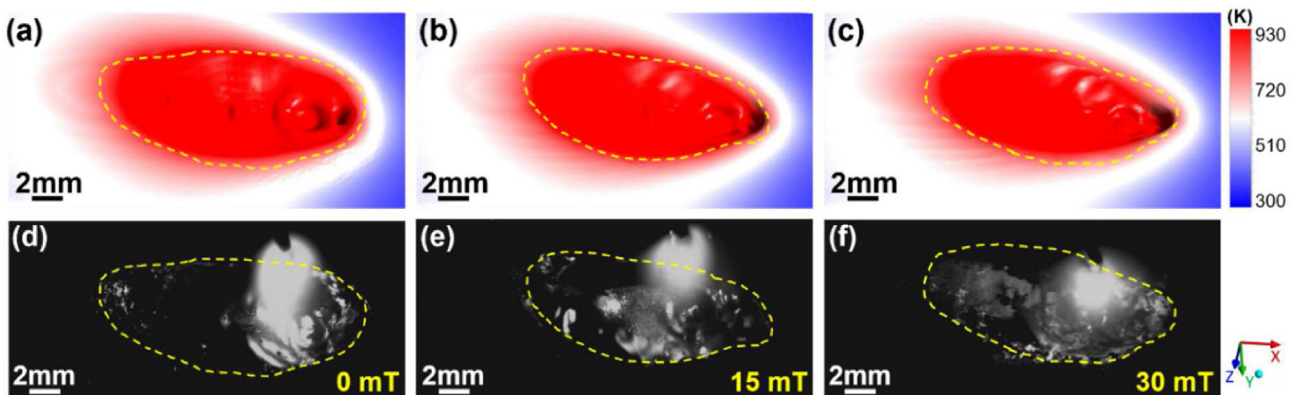


Fig. 5 – Comparison of top weld pool profiles: (a) (d) 0 mT; (b) (e) 15 mT; (c) (f) 30 mT.

may be owing to ignoring the effect of laser energy and laser-induced plasma in this model. The absorption and allocation behavior of the laser energy by plasmas can be changed because the Lorentz force created through the transverse magnetic field affect the charged particles in the plasma.

To further validate the model, the top weld pool profiles at different magnetic flux density are observed in the experiment and compared with the numerical results. The angle between the high-speed camera and the metal surface is set to 75° . In order to avoid the observation error caused by the angle deviation, the observation angle of the model is adjusted correspondingly. Fig. 5 reveals all the numerical results of top weld pool profiles are agreement with the actual images, which is z3. Therefore, the model in this research is proven to be reasonable.

4.2. Evolution of weld pool profiles

A 3D view of droplet, weld pool, and keyhole in LAHW obtained from the simulation model is shown in Fig. 6(a). The liquid temperature of the 6082 aluminum alloy (930 K) is set as the maximum temperature. To better observe the weld pool evolution behaviors, Fig. 6(b) suggests the weld pool surface and Fig. 6(c) suggests the central section named the longitudinal section.

Fig. 7 presents the evolution of weld pool surface in LAHW with the variation of magnetic flux density. In the absence of a magnetic field as suggested in Fig. 7(a)-7(e), the keyhole is occurred through the laser beam acts on base metal, the and the liquid metal is observed around the keyhole owing to the heat conduction. In addition, under the effect of the arc heat source, metal melting occurs in the area away from the

keyhole. Over time, the droplets are periodically transferred to the weld pool, and the heat accumulation from the droplets results in the expansion of the weld pool size. When the time increases to 228 ms corresponding to Fig. 7(e), the weld pool profile keeps a quasi-steady state, and a “melon seeds” shape is presented on the top surface. Similar evolutions are observed with the addition of the magnetic field. However, the weld pool width decreases as the magnetic flux density increases. This phenomenon is more pronounced in the front weld pool, which can be explained in Section 4.3. At 228 ms corresponding to the quasi-steady state displayed in Fig. 7(o), the top weld pool morphology gradually develops to an “ellipse” shape.

The evolution of the longitudinal section in LAHW with the variation of magnetic flux density is suggested in Fig. 8. Similar weld pool evolution behaviors on the longitudinal section are obtained as the magnetic flux density increases. A narrow and deep keyhole appeared and the keyhole depth is gradually increased over time. Furthermore, the droplet transfers into the weld pool, which increases the heat accumulation and expand the weld pool area. When the magnetic field is not employed, the fluctuation under the effect of the impact force caused by the dropout is occurred on the top surface of the weld pool, which can be obviously observed in the quasi-steady state in Fig. 8(e). The introduction of the external magnetic field leads to a decrease in the amplitude of the fluctuations on the weld pool. When the magnetic flux density is increased to 30 mT, the fluctuations nearly disappear at 124 ms and the weld pool profiles remain in a steady state. Figs. 8(e) and 8(j), and 8(o) suggest the morphology of the weld pool at quasi-steady state of 228 ms, it can be found that the weld pool height is significantly improved when the

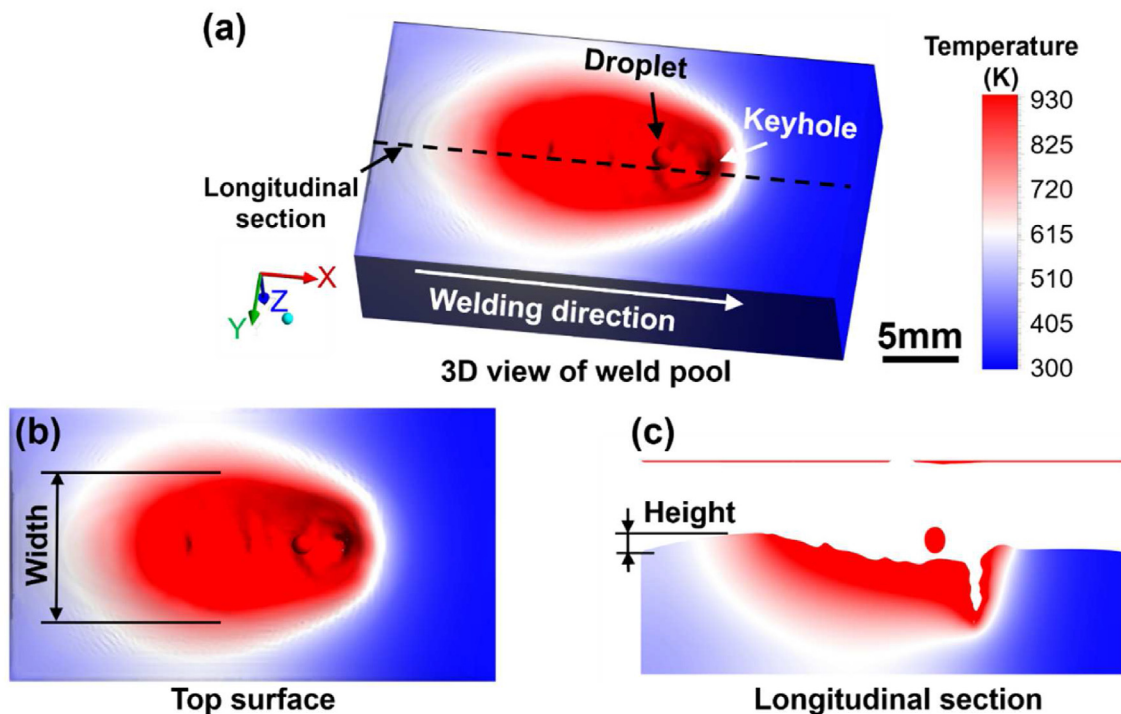


Fig. 6 – Weld pool dynamics: (a) 3D view of weld pool; (b) top surface; (c) longitudinal section.

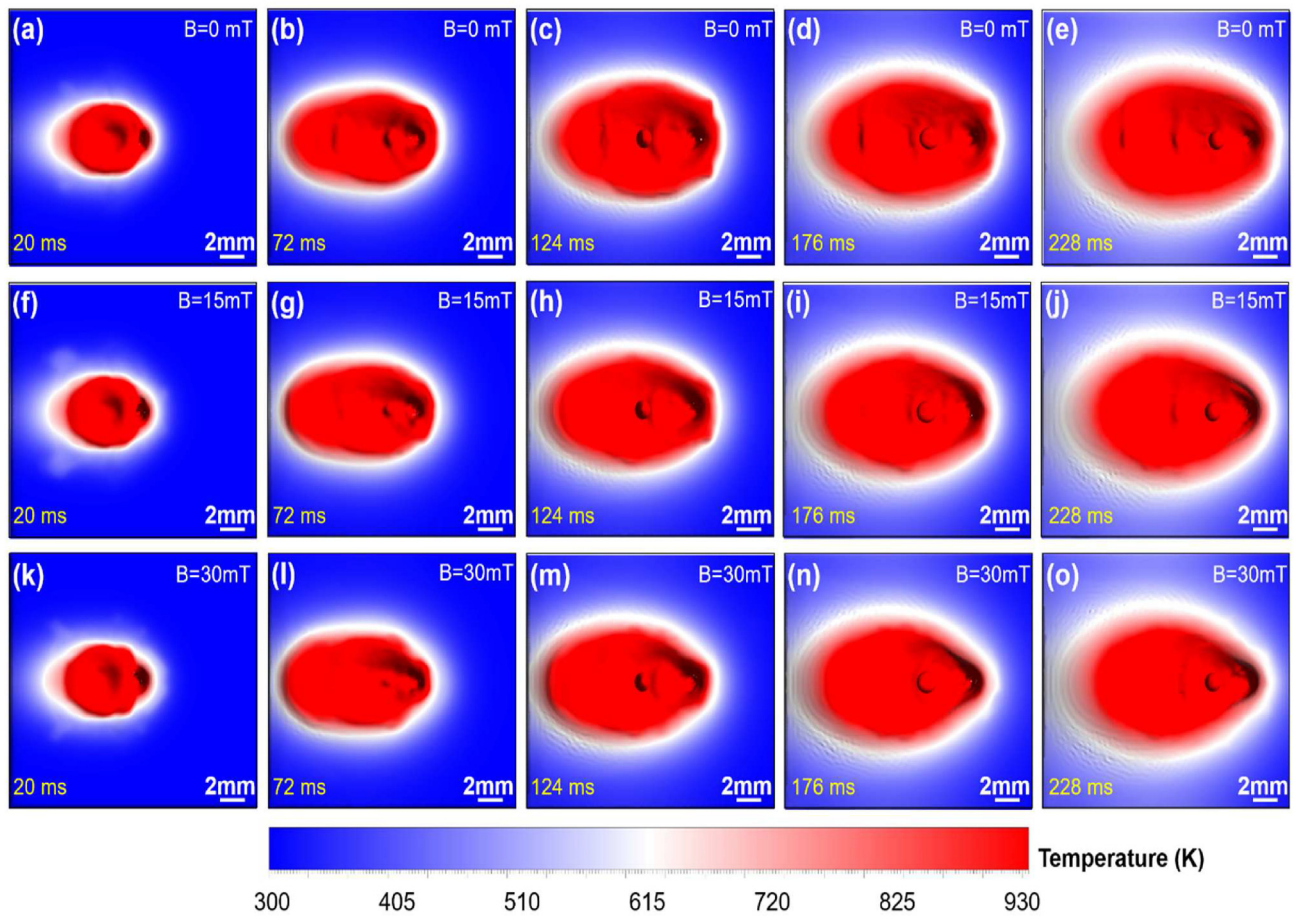


Fig. 7 – Evolution of weld pool on top surface (a)–(e) 0 mT; (f)–(j) 15 mT; (k)–(o) 30 mT.

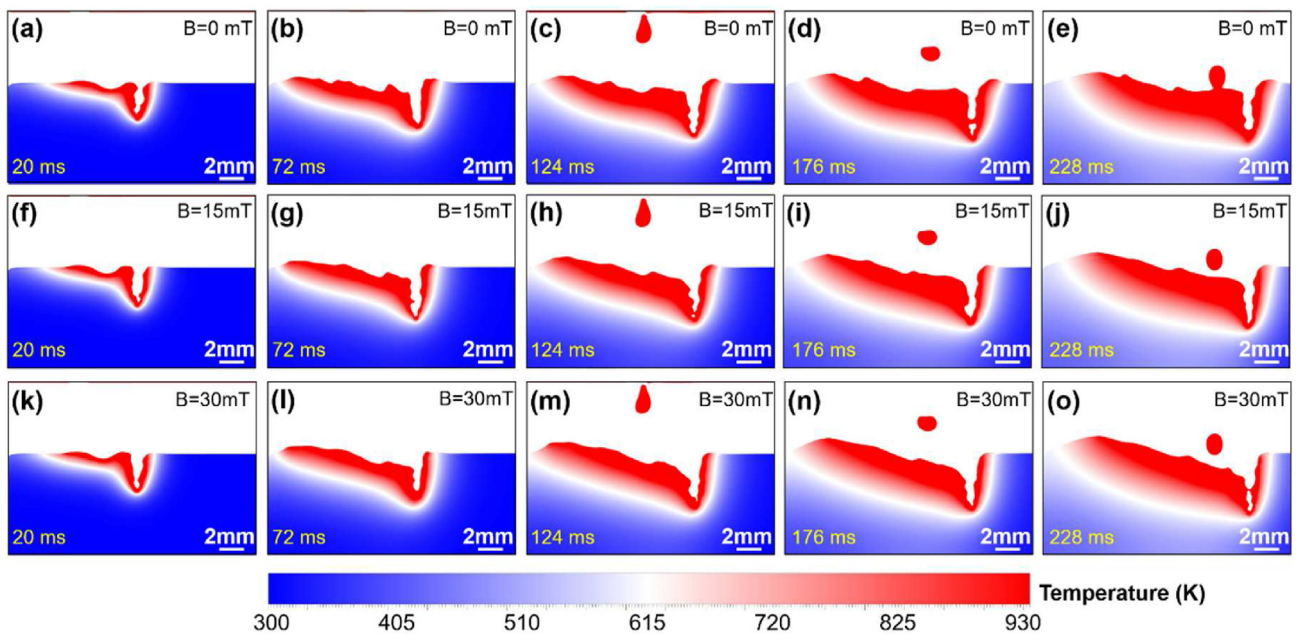


Fig. 8 – Evolution of weld pool on longitudinal section: (a)–(e) 0 mT; (f)–(j) 15 mT; (k)–(o) 30 mT.

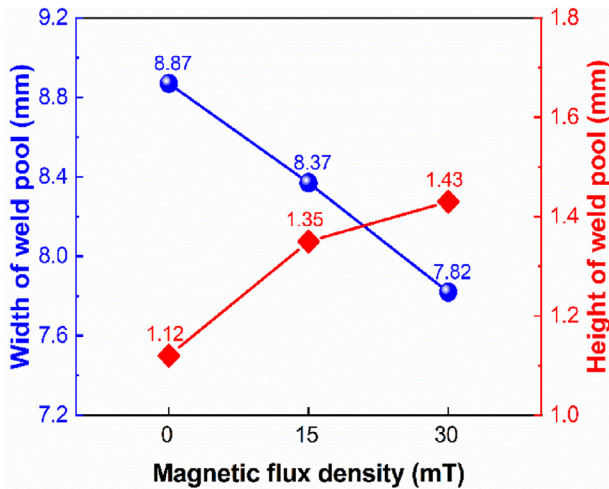


Fig. 9 – Height and width of the weld pool at 228 ms.

magnetic flux density is elevated. Moreover, the opening size of keyhole is larger, confirming that the external magnetic field is beneficial to obtain a stable weld pool.

Fig. 9 presents the variation of the weld pool height and width at quasi-steady state of 228 ms under different magnetic flux density. It indicates the magnetic field has a deep influence on the weld pool profiles. The weld pool width is reduced from 8.87 mm to 7.82 mm while its height is increased from 1.12 mm to 1.43 mm with the improvement of the magnetic field density. The heat distribution in the base metal is a crucial parameter in the weld pool profile. Moreover, the convection instead of the diffusion is more responsible for heat transfer of the weld pool because the Peclet number is greater than 200 [14]. Thus, the variation of the weld pool profiles is mainly attribute to the thermal fluid flow behavior under external magnetic field, which will be discussed in Section 4.3.

4.3. Lorentz force distribution and melts flow behavior

The Lorentz force distribution on the longitudinal section of the weld pool is displayed in Fig. 10 under a magnetic flux density of 30 mT. The weld pool profiles are divided into three areas. The Lorentz force is downward and backward in the

front parts, backward in the middle part, and upward in the rear part. Furthermore, it reaches a maximum of $1.80 \times 10^5 \text{ N/m}^3$, which is adequate to drive the melt flow. Consequently, the Lorentz force promotes the backward flow in the whole weld pool and suppresses the downward flow in the rear weld pool.

Fig. 11 suggests the melt flow behaviors in a period without the magnetic field. Before the droplet enters the weld pool at 0.528 s shown in Fig. 11(a), the strong recoil pressure and surface tension on the keyhole drive the melt at the bottom of the keyhole flow upwards from a position adjacent to the keyhole rear wall. Additionally, the melts behind the keyhole rear wall flow backward on the weld pool surface on account of the arc shear stress and the surface tension. The flowing direction is converted to be forward when the top melts are blocked by the rear periphery of weld pool. Therefore, a vortex appeared in the front weld pool due to the collision of melts. As the droplet drops into the weld pool at 0.530 s illustrated in Fig. 11(b), a strong downward impact force is presented in the top area behind the keyhole, which changes the melts flow pattern of the front weld pool. Moreover, a flowing source appears at the top area of the front weld pool owing to the impact force. The melts flow downward and forward in the front part, which inhibits the upward flow coming from the keyhole bottom and facilitates the backward flow along the bottom weld pool. Thus, the vortex is driven back into a lower part of the middle weld pool. Fig. 11(c) at 0.532 s depicts a violent oscillation formed in the front area, which is attribute to the hydrodynamic pressure converted from the impact force of the droplet. Thus, series small vortices are observed behind the keyhole. With the time evolves to 0.534 s and 0.536 s, the hydrodynamic pressure has totally replaced the impact force from the droplets, and the unstable oscillations disappear in the front weld pool and the melts flow behavior becomes more regular. Fig. 11(d) and (e) show the melts flow forward in the front weld pool, and they flow upward and backward adjacent to the keyhole rear wall. Additionally, the vortex still exists in the middle weld pool, which is attribute to the reversal of the melt flow direction.

Fig. 12 presents the melts flow behaviors on the longitudinal section in a period during LAHW under magnetic flux density of 15 mT. Some significant differences of the melts flow patterns are observed compared to the situation without magnetic field. The backward Lorentz force becomes a major driving force in the front and middle weld pool, which reverses the melts flowing direction to be backward. In the rear

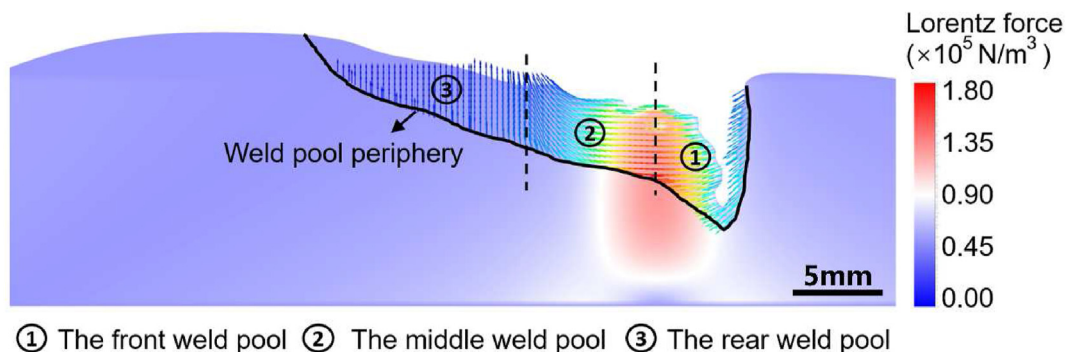


Fig. 10 – Lorentz force distribution on the longitudinal section.

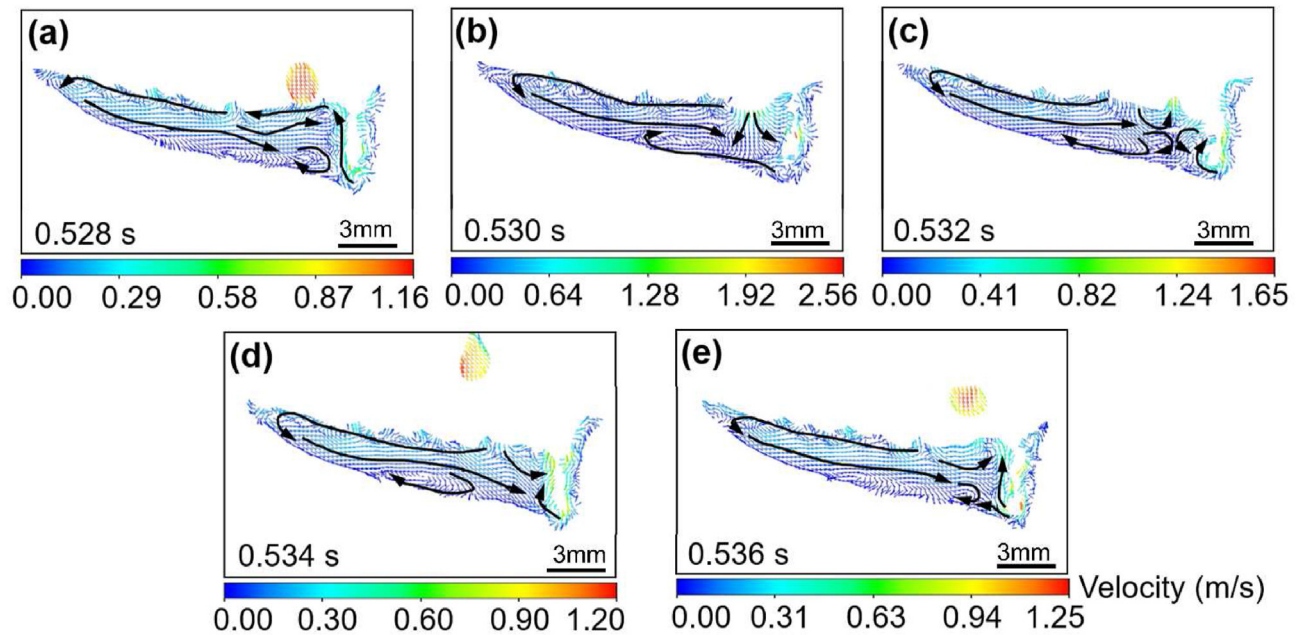


Fig. 11 – Melts flow behaviors on the longitudinal section in a period without the magnetic field: (a) 0.528 s; (b) 0.530 s; (c) 0.532 s; (d) 0.534 s; (e) 0.536 s.

weld pool, a new collision zone is observed. The melts are promoted to flow upward at the front part of the rear weld pool through the upward Lorentz force. Thus, the melts flow backward at the top area of the rear weld pool. When the flow is obstructed by the rear weld pool edge, the flowing direction is changed to be forward. Additionally, the violent collision causes the melts in the middle weld pool to flow forward. Subsequently, in Fig. 12(a), (b), and (d), the melts mentioned

above meet the melts flowing backward from the keyhole bottom, leading to the appearance of the vortex at the upper middle part instead of the lower middle part of weld pool. Furthermore, compared to Figs. 11(a) and 12(a), the dropping velocity of the droplet is reduced from 1.16 m/s under no magnetic field to 1.14 m/s at 0.528 s. The distance from the keyhole rear wall to the position where droplet transfers into the weld pool surface is also increased. It reveals that the

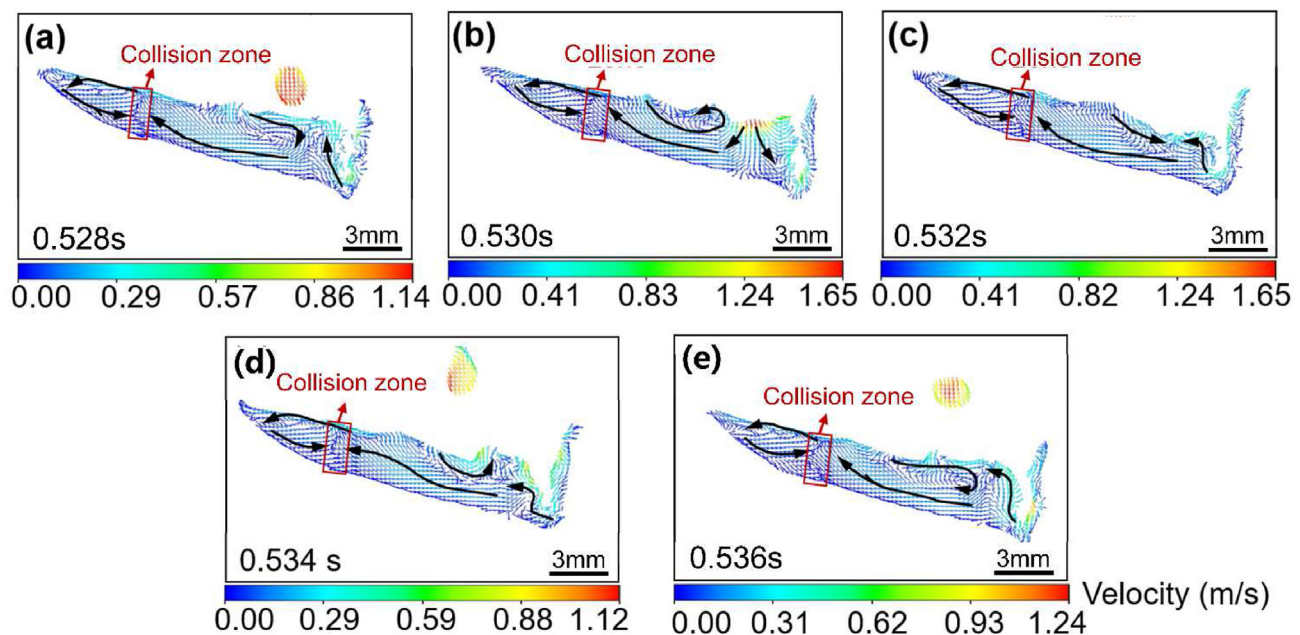


Fig. 12 – Melts flow behaviors on the longitudinal section in a period with 15 mT: (a) 0.528 s; (b) 0.530 s; (c) 0.532 s; (d) 0.534 s; (e) 0.536 s.

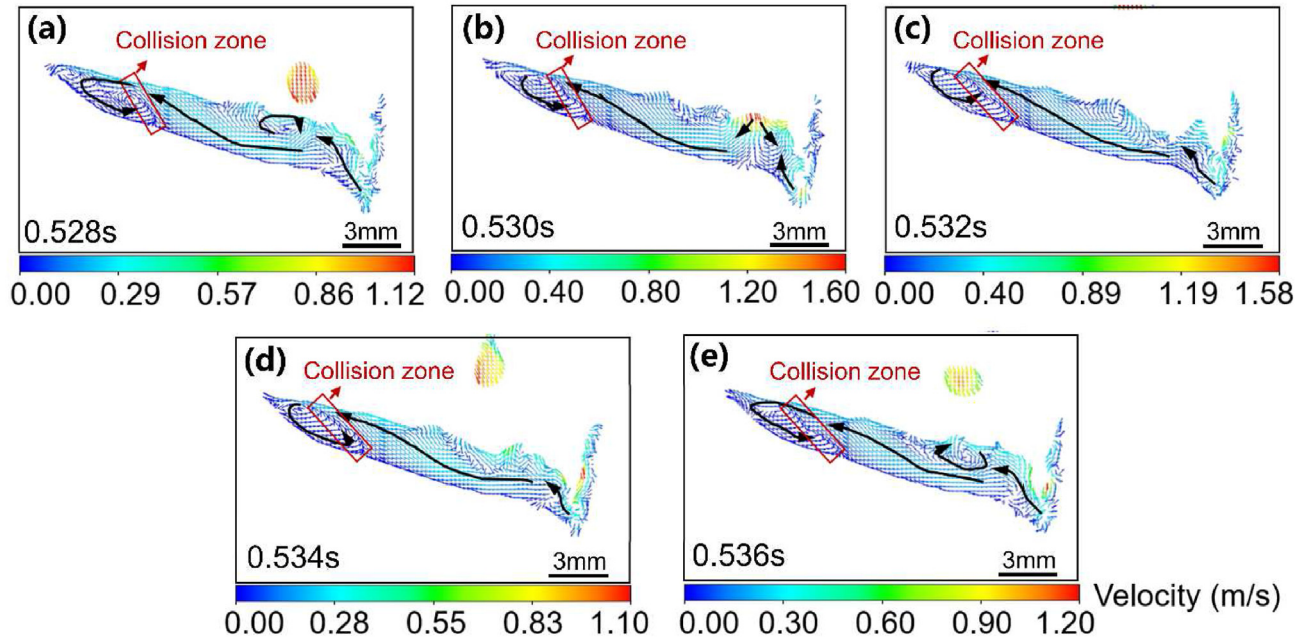


Fig. 13 – Melts flow behaviors on the longitudinal section in a period with 30 mT: (a) 0.528 s; (b) 0.530 s; (c) 0.532 s; (d) 0.534 s; (e) 0.536 s.

influence of the droplet is weakened, resulting in the enhancement of the weld pool stability. Moreover, compared to the moment of the droplet transfers to the weld pool suggested in Fig. 11(c) and 12(c), the forward flow in the front weld pool owing to the impact force is also suppressed by the backward Lorentz force, leading to a decrease of small vortex behind the keyhole rear wall.

Fig. 13 shows the melts flow behaviors in a period during LAHW when magnetic flux density is 30 mT. The improvement of the magnetic flux density causes a stronger Lorentz force, which further facilitates the backward flow in the middle part and upward flow in the rear weld pool. Moreover, the backward flowing velocity is obviously increased, and the region where the melts flow forward in the rear weld pool is

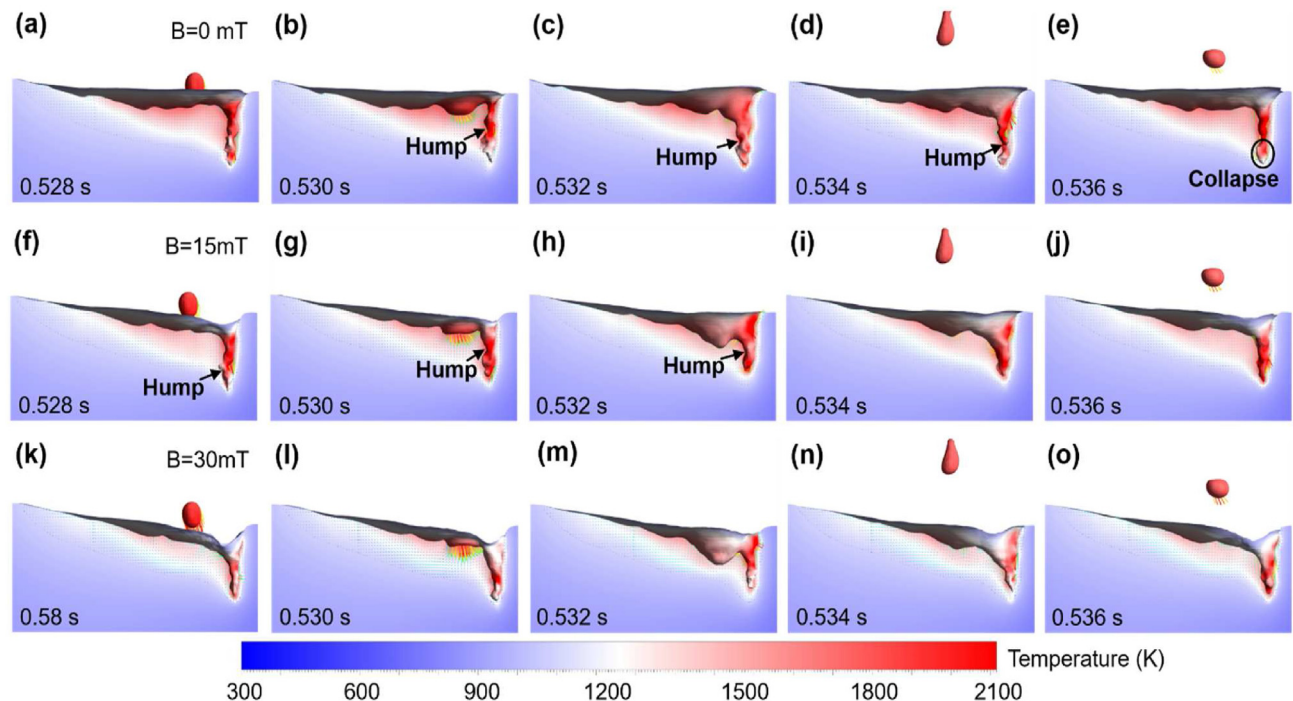


Fig. 14 – Keyhole dynamics in a period during LAHW: (a)–(e) 0 mT; (f)–(j) 15 mT; (k)–(o) 30 mT.

shorten. Consequently, the collision zone is pushed backward. Furthermore, the dropping velocity of the droplet is further decreased to 1.12 m/s at the 0.528 s, and the distance from the keyhole to the point where the droplet transfers to the weld pool is further extended. The stronger backward Lorentz force further suppresses the forward flow in the front weld pool, which accelerates the backward flow of melts originating from the keyhole.

In the LAHW process, the heat inputs contain the arc energy, laser energy, and the heat from the droplet. The arc energy is directly applied to the surface, while the laser energy is acted on the keyhole wall. Additionally, the heat from the droplet is also transferred into the weld pool. Thus, the equivalent heat source contains the arc area on the top surface, the keyhole wall, and the droplet. When the magnetic field is used, the arc deflects backward and more energy distributes backward in the weld pool, causing the variation of the arc source distribution coefficient. Furthermore, the backward component of the Lorentz force promotes the backward flow in the whole weld pool. Thus, the heat accumulation of the weld pool sides becomes less, leading to the decrease of the weld width in the front weld pool. Moreover, the upward Lorentz force makes the melts in the rear weld pool reverse to flowing upward. More heat is absorbed from the walls of the keyhole and the melt droplet, which leads to an increase of the weld pool width.

4.4. Keyhole dynamics

The keyhole dynamics during the LAHW process are displayed in Fig. 14. The keyhole shape is slender and keyhole is in dynamic state when no magnetic field is employed. The keyhole shrinks and oscillates in the width direction during the whole welding process. Fig. 14(b) and (c) and illustrate the process where the weld pool is impinged by the droplet. The melts are

squeezed to the keyhole rear wall by the strong impact force of droplet, resulting in the generation of the keyhole humps. Fig. 14(d) show the results after the entry of the droplet. Moreover, the melt in the post weld pool flows forwards to the keyhole rear wall with the action of the oscillating hydrodynamic pressure, which also produces humps on the keyhole rear wall. The keyhole humps keep growing forward. When the keyhole humps impact the keyhole front wall, the keyhole collapse occurs, as presented in Fig. 14(e). These phenomena reveal that the keyhole stability in the LAHW process without the magnetic field is extremely poor.

The backward Lorentz force behind the keyhole rear wall decreases the forward flow produced by droplet impact and vortex under the magnetic field, which reduces the local deformation of the keyhole rear wall. Moreover, the strong Lorentz force also promotes the backward flow of the melts in the front and middle weld pool. It is the equivalent of applying a backward tension acting on the keyhole, which increases the keyhole width. Fig. 15 illustrates the keyhole width at the middle keyhole depth under different magnetic flux density. The average keyhole width without the magnetic field is 0.74 mm. However, it is improved to 0.76 mm with the assistance of a magnetic flux density of 15 mT, and further increased to 0.83 mm corresponding to the 30 mT. Increase of the keyhole width is favorable for facilitating the escape of metal vapor, which also reduces the keyhole fluctuation [38]. Furthermore, a wider keyhole makes it more difficult to connect the hump to the keyhole front wall, which improves the keyhole capillary stability. Thus, fewer humps are observed when the magnetic flux density is 15 mT, as shown in Fig. 14(f)–14(j). Furthermore, in Fig. 14(k)–14(l) corresponding to 30 mT, no obvious hump is found, and keyhole stability is further improved.

To verify the effect of steady magnetic field on the keyhole stability, Fig. 16(a) collects the keyhole depth in LAHW process

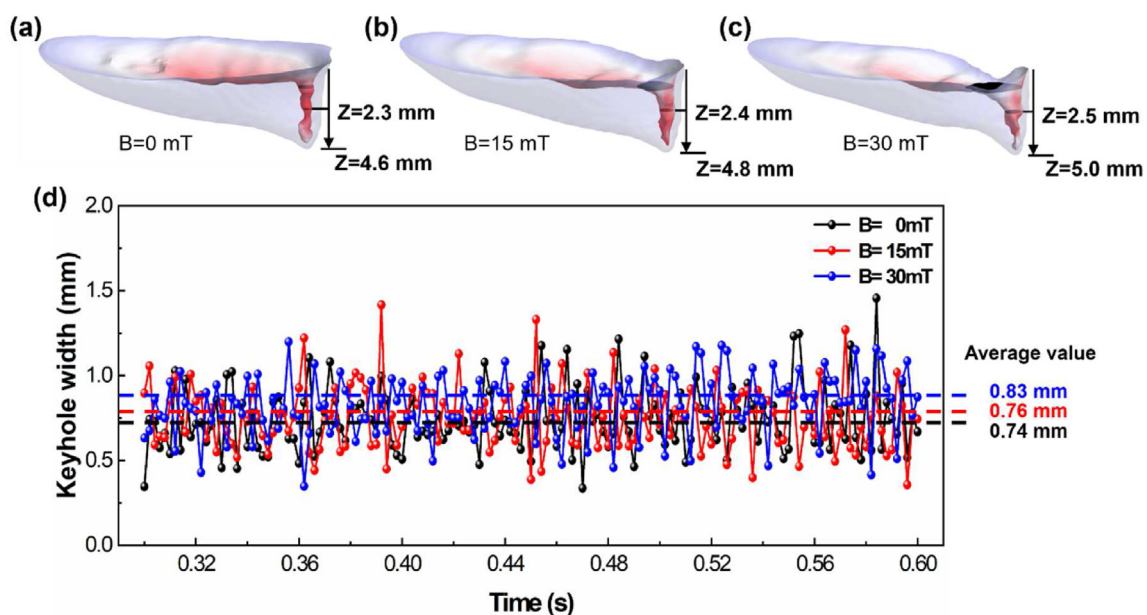


Fig. 15 – Comparison of the keyhole width: (a) section position under 0 mT; (b) section position under 15 mT; (c) section position under 30 mT; (d) curve of keyhole width variation.

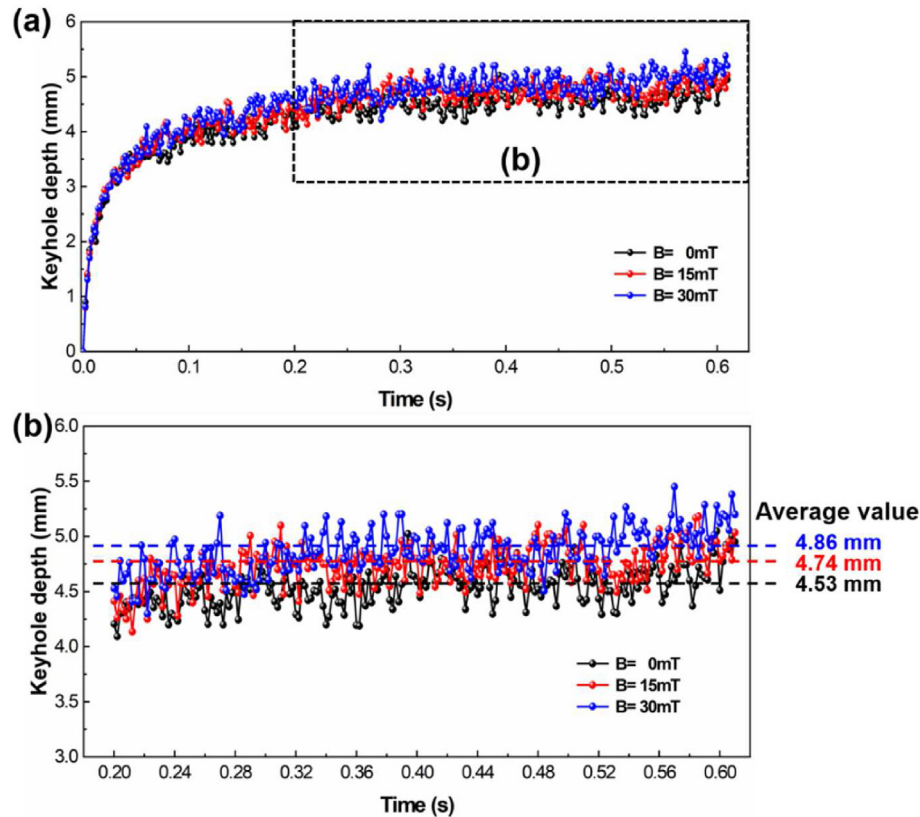


Fig. 16 – Curve of keyhole depth versus time in LAHW: (a) in the whole welding process: (b) at the quasi-steady state.

with the variation of the magnetic flux density. It can be found that the keyhole depth is fluctuating in the whole welding process. The keyhole depth is increased at the initial stage and its variation is at quasi-steady state after 0.20 s. The average values of keyhole depths at quasi-steady state were displayed in Fig. 16(b). It is measured 4.53 mm without the magnetic field. As the magnetic flux density increases to 15 mT and 30 mT, the average is increased to 4.74 mm and 4.86 mm. Increase of keyhole depth represents that more laser energy can be directly absorbed at the keyhole bottom, which confirms that the magnetic field plays an active role in improving keyhole stability.

4.5. Comparative analysis with the case in MFLBW

The study on weld pool behaviors and keyhole dynamics in MFA-LBW has been conducted using the similar experimental setup and numerical analysis in our previous literature [39]. To deepen the physics understanding of MFA-LAHW process, the difference on physical behaviors including Lorentz force distribution, weld pool behaviors, and keyhole dynamics between MFA-LAHW and MFA-LBW is comparatively analyzed, which is listed in Table 4. Generally, the welding current in LAHW of the medium-thickness aluminum alloy is between 100 and 300 A and its current density is the order of magnitude

Table 4 – Comparison between MFA-LAHW and MFA-LBW.

	MFA-LAHW	MFA-LBW
Magnetic flux density	$\sim 10^{-2}$ T	$\sim 10^{-1}$ T
Current density type interacting with magnetic field	External welding current	Self-induced current
Lorentz force distribution	Backward and downward in front part Backward in middle part Upward in rear part	Upward in upper region Backward in middle region Downward in lower region
Melts flow behaviors in longitudinal section	Backward flow is promoted Forward flow is suppressed	Backward flow is suppressed Forward flow is suppressed
Weld pool width	Reduced	Reduced
Weld pool length	Increased	Reduced
keyhole depth	Increased	Reduced
Keyhole stability	Improved	Improved

of 10^7 A/m². Consequently, only a lower magnetic flux density with the order of magnitude of 10^{-2} T interacts the current density to produce a Lorentz force the order of magnitude of 10^5 A/m², which has been proven to be adequate to drive the melts flow. However, For MFA-LBW, the self-induced current density is generated in the weld pool when external magnetic field is employed to the flowing conductive fluid. The self-induced current density is displayed as follows:

$$\vec{j} = \sigma_e (\vec{v} \times \vec{B}_0 + \vec{E}) \quad (29)$$

where, σ_e represents electrical conductivity, $\vec{v}$ is flow velocity, $\vec{B}_0$ represents the magnetic flux density, and $\vec{E}$ represents the induced electric field. The strength of Lorentz force is determined by the magnitudes of magnetic flux density and melts flow velocity because it was produced through the interaction between the magnetic field and its self-induced current density. The melts flow velocity in LBW is the magnitude of 10^{-1} m/s. As a result, a higher magnetic flux density ($\sim 10^{-1}$ T) is used to generate a sufficient Lorentz force ($\sim 10^5$ A/m²).

When the same external magnetic field arrangements are employed in LAHW and LBW of medium-thickness aluminum alloy plate, the obvious different Lorentz force distribution and melts flow behaviors in the weld pool are obtained, which is attributed to different sources of current density as mentioned above. Thus, the keyhole dynamics and the weld pool profiles have the different variation trends with addition of magnetic field, which affects the weld formation and defects suppression. The porosity defect is a troubling problem for deep-penetration LBW or LAHW of medium-thickness aluminum alloy. Stabilizing the keyhole and facilitating the escape of keyhole-induced bubbles are two main factors for suppressing porosity defects. Therefore, it will be more effective and valuable to weld medium-thickness aluminum alloy by using MFA-LAHW.

5. Conclusions

This research established a numerical model to investigate the keyhole dynamics and weld pool behaviors in MFA-LAHW. The conclusions can be summarized as follows.

- (1) Different from the case in MFA-LBW, the presence of arc induces the interaction between welding current and external magnetic field with a lower magnetic flux density (≤ 30 mT), resulting in a strong Lorentz force up to 1.80×10^5 N/m³, which is adequate to drive melts flow behaviors.
- (2) The weld pool profile is optimized by the magnetic field. The backward flow of melts in the front and middle weld pool is facilitated and the heat accumulation is reduced, leading to the decrease of weld pool width. Furthermore, the weld pool height is increased because the upward Lorentz force converts the thermal fluid flowing direction to be upward in the rear weld pool.
- (3) The keyhole stability is improved by external magnetic field. The forward flow produced by droplet impact and the vortex is suppressed, decreasing the keyhole humps. Moreover, as the magnetic flux density

increases from 0 mT to 30 mT, the average keyhole width increases from 0.74 mm to 0.83 mm, enhancing the stability of the capillary formed by the keyhole.

- (4) For eliminating porosity defects in medium-thickness aluminum alloy, it may be more effective for MFA-LAHW compared to MFA-LBW. A magnetic field with the lower magnetic flux density ($\sim 10^{-2}$ T) can stabilize the keyhole and induce backward flow, which is conducive to the reduction of the porosity defects.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this study.

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REFERENCES

- [1] Bagger C, Olsen FO. Review of laser hybrid welding. *J Laser Appl* 2005;17(1):2–14.
- [2] Steen W. Arc augmented laser processing of materials. *J Appl Phys* 1980;51(11):5636–41.
- [3] Zhang C, Gao M, Wang D, Yin J, Zeng X. Relationship between pool characteristic and weld porosity in laser arc hybrid welding of AA6082 aluminum alloy. *J Mater Process Technol* 2017;240:217–22.
- [4] Nomura K, Ogino Y, Hirata Y. Shape control of TIG arc plasma by cusp type magnetic field with permanent magnets. *Weld Int* 2012;26(10):759–64.
- [5] Shoichi M, Yukio M, Koki T, Yasushi T, Yukinori M, Yusuke M. Study on the application for electromagnetic controlled molten pool welding process in overhead and flat position welding. *Sci Technol Weld Join* 2013;18(1):38–44.
- [6] Kern M, Berger P, Hügel H. Magneto-fluid dynamic control of seam quality in CO₂ laser beam welding. *Weld J* 2000;79:72–8.
- [7] Avilov V, Fritzsche A, Bachmann M, Gumenyuk A, Rethmeier M. Full penetration laser beam welding of thick duplex steel plates with electromagnetic weld pool support. *J Laser Appl* 2015;28(2):022420.
- [8] Fritzsche A, Hilgenberg K, Teichmann F, Pries H, Dilger K, Rethmeier M. Improved degassing in laser beam welding of aluminum die casting by an electromagnetic field. *J Mater Process Technol* 2018;253:51–6.
- [9] Gatzert M. Influence of low-frequency magnetic fields during laser beam welding of aluminium with filler wire. *Phys Procedia* 2012;39:59–66.
- [10] Meng X, Bachmann M, Artinov A, Rethmeier M. The influence of magnetic field orientation on metal mixing in electromagnetic stirring enhanced wire feed laser beam welding. *J Mater Process Technol* 2021;294:117135.

- [11] Chen R, Kong H, Luan J, Wang A, Liu C. Effect of external applied magnetic field on microstructures and mechanical properties of laser welding joint of medium-Mn nanostructured steel. *Mater Sci Eng* 2020;792:139787.
- [12] Hu C, Yan F, Zhu Z, Xu Y, Wang C. Effects on microstructural refinement of mechanical properties in steelcopper joints laser welded with alternating magnetic field augmentation. *Mater Char* 2021;175(7):111059.
- [13] Bachmann M, Avilov V, Gumenyuk A, Rethmeier M. Numerical simulation of full-penetration laser beam welding of thick aluminium plates with inductive support. *J Phys Appl Phys* 2012;45:035201.
- [14] Bachmann M, Avilov V, Gumenyuk A, Rethmeier M. About the influence of a steady magnetic field on weld pool dynamics in partial penetration high power laser beam welding of thick aluminium parts. *Int J Heat Mass Tran* 2013;60:309–21.
- [15] Bachmann M, Avilov V, Gumenyuk A, Rethmeier M. Numerical assessment and experimental verification of the influence of the Hartmann effect in laser beam welding processes by steady magnetic fields. *Int J Therm Sci* 2016;101:24–34.
- [16] Cao L, Shao X, Jiang P, Zhou Q, Rong Y, Geng S, et al. Effects of welding speed on microstructure and mechanical property of fiber laser welded dissimilar butt joints between AISI316L and EH36. *Metals* 2017;7(7):270.
- [17] Zhang R, Tang X, Xu L, Lu F, Cui H. Mechanism study of thermal fluid flow and weld root hump suppression in full penetration laser welding of Al alloy with alternating magnetic field support. *Int J Heat Mass Tran* 2021;166:120759.
- [18] Liu F, Wang H, Meng X, Tan C, Chen B, Song X. Effect of magnetic field orientation on suppressing porosity in steady-magnetic-field-assisted aluminum alloy deep-penetration laser welding. *J Mater Process Technol* 2022;304:117569.
- [19] Meng X, Artinov A, Bachmann M, Rethmeier M. Numerical and experimental investigation of thermo-fluid flow and element transport in electromagnetic stirring enhanced wire feed laser beam welding. *Int J Heat Mass Tran* 2019;144:118663.
- [20] Meng X, Artinov A, Bachmann M, Üstündağ Ö, Gumenyuk A, Rethmeier M. The detrimental molten pool narrowing phenomenon in wire feed laser beam welding and its suppression by magnetohydrodynamic technique. *Int J Heat Mass Tran* 2022;193:122913.
- [21] Chen R, Jiang P, Shao X, Mi G, Wang C. Effect of magnetic field applied during laser-arc hybrid welding in improving the pitting resistance of the welded zone in austenitic stainless steel. *Corrosion Sci* 2017;126:385–91.
- [22] Chen R, Jiang P, Shao X, Mi G, Wang C. Enhancement of fatigue resistance for 316L welds produced by magnetic field assisted laser-MIG hybrid welding. *J Mater Process Technol* 2018;254:114–23.
- [23] Chen R, Jiang P, Shao X, Mi G, Wang C, Geng S, et al. Improvement of low-temperature impact toughness for 304 weld joint produced by laser-MIG hybrid welding under magnetic field. *J Mater Process Technol* 2017;247:306–14.
- [24] Liu F, Tan C, Wu L, Gong X, Chen B, Song X, et al. Influence of waveforms on Laser-MIG hybrid welding characteristics of 5052 aluminum alloy assisted by magnetic field. *Opt Laser Technol* 2020;132:106508.
- [25] Xu B, Liu F, Lin D, Han X, Chen X, Wu L, et al. Investigation of weld formation and porosity in 5052 aluminium alloy laser-GMA hybrid welding assisted by magnetic field with different orientations. *Sci Technol Weld Join* 2022;27(8):672–82.
- [26] Goldak J, Chakravarti A, Bibby M. A new finite element model for welding heat sources. *Metall Trans A B* 1984;15(2):299–305.
- [27] Gao X, Wu C, Goecke S, Kügler H. Numerical simulation of temperature field, fluid flow and weld bead formation in oscillating single mode laser-GMA hybrid welding. *J Mater Process Technol* 2017;242:147–59.
- [28] Zhou J, Tsai H. Modeling of transport phenomena in hybrid laser-MIG keyhole welding. *Int J Heat Mass Tran* 2008;51(17–18):4353–66.
- [29] Meng X, Qin G, Zou Z. Investigation of humping defect in high speed gas tungsten arc welding by numerical modelling. *Mater Des* 2016;94:69–78.
- [30] Campbell S, Galloway A, McPherson N. Arc pressure and fluid flow during alternating shielding gases. Part 2: arc force determination. *Sci Technol Weld Join* 2013;18(7):597–602.
- [31] Wu C. Fluid flow and heat transfer in GMA weld pools. *Weld J* 1988;67(3):70s–5s.
- [32] Brackbill J, Kothe D, Zemach C. A continuum method for modeling surface tension. *J Comput Phys* 1992;100(2):335–54.
- [33] Amara E, Fabbro R. Modelling of gas jet effect on the melt pool movements during deep penetration laser welding. *J Phys D Appl Phys* 2008;41(5):055503.
- [34] Ki H, Mazumder J, Mohanty P. Modeling of laser keyhole welding: Part II. simulation of keyhole evolution, velocity, temperature profile, and experimental verification. *Metall Mater Trans* 2002;33(6):1831–42.
- [35] Oreper G, Szekely J. Heat- and fluid-flow phenomena in weld pools. *J Fluid Mech* 2006;147:53–79.
- [36] Li Y, Wu C, Wang L, Gao J. Analysis of additional electromagnetic force for mitigating the humping bead in high-speed gas metal arc welding. *J Mater Process Technol* 2016;229:207–15.
- [37] Wang L, Chen J, Wu C, Luan S. Numerical analysis of arc and droplet behaviors in gas metal arc welding with external compound magnetic field. *J Mater Process Technol* 2020;282:116638.
- [38] Geng S, Yang W, Jiang P, Han C, Ren L. Numerical study of keyhole dynamics and porosity formation during high-power oscillating laser welding of medium-thick aluminum alloy plates. *Int J Heat Mass Tran* 2022;194:123084.
- [39] Tan C, Liu Y, Xu B, Wang H, Liu F, Gong X, et al. Numerical and experimental study of thermal fluid flow and keyhole dynamic in laser welding of aluminum alloy assisted by electromagnetic field. *Opt Laser Technol* 2023;157:108718.

FURTHER READING

- [40] Liu F, Xu B, Song K, Tan C, Zhao H, Wang G, et al. Improvement of penetration ability of heat source for 316 stainless steel welds produced by alternating magnetic field assisted laser-MIG hybrid welding. *J Mater Process Technol* 2022;299:117329.